Khue (Cherry) Pham

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Education

Northeastern University, The Roux Institute – M.S. Computer Science Dec 2027

Relevant Coursework: Programming Design Paradigms; Machine Learning; Database Management Systems

Olin College of Engineering – B.S. Electrical and Computer Engineering May 2025

Technical Skills

Languages: Python, SQL

Software Engineering: AWS, Git

ML/AI: Hugging Face Transformers, LangChain, PyTorch, LLMs, Fine-tuning, RAG, NLP, Prompt Engineering, ONNX, semantic search, LLaMA, Quantization

Experience

Florida Center for Reading Research Remote

Machine Learning Engineer Sep 2024 – Aug 2025

- Built a PyTorch/LangChain pipeline and fine-tuned a LLaMA-based LLM on early literacy corpora, evaluating prompting and fine-tuning variants against rubric-aligned benchmarks to select version for cross-state pilot deployment.
- Built RAG pipelines with pgvector-backed semantic search (PostgreSQL), improving retrieval accuracy by 15–25% on children's reading development tasks; evaluated using BLEU/ROUGE benchmarks.
- Deployed LLM inference behind a REST API endpoint serving an education portal, with scalable cloud infrastructure supporting SQL/NoSQL storage and OAuth-based authentication for multi-user research access.

Air Partners Needham, MA

Data Engineer Jun 2023 – Aug 2023

- Designed a normalized data model and modular ETL pipeline to integrate time-series data from 18 disparate IoT sensors into an analysis-ready dataset, processing ≈3 million records across 2 years of hourly readings; reduced pipeline latency by 30% through parallelized processing and optimized API batching.

Olin College – College as a Living Lab Needham, MA

AI Pedagogy Researcher Jun 2025 – Aug 2025

- Designed and deployed a GPT-4o-based automated grading pipeline integrated with the Canvas LMS API, processing 500–1,000 student submissions per semester with tokenization and semantic similarity analysis.
- Evaluated prompting strategies across 4–6 model iterations against professor-graded rubrics, reaching 80–87% alignment with human graders through iterative prompt design and layer-specific fine-tuning.

Projects

ENCoM – Real-Time Generative Speech-to-Speech System | GitHub

- Trained a Cross-Modal Adapter to align Mimi audio latent spaces with a pretrained SmolLM2-360M language model, preserving >95% of reasoning capabilities while minimizing VRAM footprint for edge deployment.
- Designed and ran experiments comparing a Mimi+Mamba hybrid architecture against a controlled cascaded baseline, achieving 2.79x inference speedup while maintaining audio quality (MOS) on Switchboard/Expresso evaluation sets.
- Engineered a low-latency pipeline integrating the Mimi neural audio codec with a Mamba SSM backbone and Temporal/Depth Transformer, reducing time-to-first-audio to <300ms.

Next-Step – Full-Stack School Management Platform | GitHub

- Designed a cross-platform REST API with role-based access control for 3 user roles using async Python (FastAPI, SQLAlchemy 2.0).
- Integrated Google Drive API and OAuth 2.0 authentication pipeline enabling secure file storage, permission management, and real-time data synchronization across iOS, Android, and Web clients.
- Containerized with Docker and deployed via multi-worker uvicorn, validated at 100 concurrent users with 7,801 requests served, zero errors, and 95% of responses under 200ms.